

ERP User Manual Training Guide

op portal, mobile warehouse app, procurement,
ufacturing, inventory, sales, and administration — for
visors, warehouse, procurement, billing, and admin users.

1.0

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Portal: ERP Portal (desktop) + Mobile Warehouse PWA

Audience: Supervisors, procurement, warehouse staff, billing, and administrators

Table of contents

1. Introduction
 2. Getting started
 3. Roles and access
 4. Physical layout and terminology
 5. Desktop navigation
 6. Dashboard
 7. Products and catalog
 8. Procurement
 9. Inventory and stock
 10. Warehouse and locations
 11. Transfer orders
 12. Manufacturing
 13. Sales, quotes, and fulfilment
 14. Billing and POS
 15. Transport and dispatch
 16. Returns and credit notes
 17. Approvals
 18. Reports
 19. Settings and administration
 20. Mobile warehouse app
 21. Automation: stock rules
 22. End-to-end scenarios
 23. Glossary
 24. Tips and troubleshooting
-

1. Introduction

PVS ERP is an integrated system for a food-processing plant. It connects:

- **Buying** raw materials (procurement, GRN)
- **Making** products (manufacturing orders, BOMs, work orders)
- **Storing** stock (warehouses, bins, transfers)
- **Selling** to customers (quotes, sales orders, invoices)
- **Shipping** (pick, pack, trips)

The same stock ledger powers desktop screens and the **mobile warehouse app** used on the shop floor with barcode scanning.

Two applications

Application	URL (dev)	Who uses it
ERP Portal (desktop)	<code>http://localhost:5173/</code>	Office, supervisors, procurement, billing
Mobile PWA	<code>/m/</code> on same host	Warehouse workers, floor staff

Workers typically log in on mobile with a **6-digit PIN**. Office staff use **username + password**.

2. Getting started

2.1 Logging in (desktop)

1. Open the ERP Portal.
2. Enter **username** and **password**.
3. You land on the **Dashboard**.

Your role controls which menu items you see. If a module is missing from the left menu, your account does not have access — ask an administrator.

2.2 Workspace tabs

Each item you open from the left navigation opens in a **tab** at the top. You can keep several modules open (e.g. Manufacturing + Inventory) and switch between tabs without losing your place.

2.3 Command Palette (power users)

Press **Ctrl+K** or **Ctrl+I** to open the Command Palette.

Use it to:

- **Jump** to any module (Dashboard, Products, Procurement, Manufacturing, ...)
- **Search** products, invoices, vendors, workers, production orders
- **Quick actions:** New Invoice, New Purchase Order, Stock Transfer, New Production Order

2.4 Barcode scanner overlay

Press **Ctrl+B** from the desktop top bar to open the scanner overlay for quick product or location lookup.

2.5 Signing out

Use **Sign out** in the top command bar. On mobile, use **Profile** → **Sign out**.

3. Roles and access

Role	Typical users	Main modules
Admin	IT, plant manager	Everything including Settings
Supervisor	Production/plant supervisor	Manufacturing, BOMs, Productivity, most ops, Approvals
Procurement	Purchase team	Products, Procurement, Price lists, Inventory (read/adjust)
Billing	Sales counter, accounts	Customers, Quotes, SOs, Billing, Returns, Picking/Packing (office view)
Warehouse	Storekeepers	Inventory, Warehouse, Transfers, Picking, Packing, WH Audit, mobile
Worker	Floor operators	Mobile app only — tasks, scan, count

Admin bypasses all restrictions. Other roles see only the nav items listed in the system configuration.

4. Physical layout and terminology

Understanding *where* stock lives is essential before using inventory or manufacturing modules.

4.1 Warehouse types

Type	Examples	Purpose
Storage (godowns)	WH-STOR (Big Godown), WH-GDNW (New Godown), WH-STO-OILSEEDS, WH-RAW	Long-term bulk raw materials
Production	WH-PROD-OIL, WH-PROD-MILL, WH-PROD-SOAP, ...	Shop-floor buffer: materials staged here before/during MO; temporary FG before putaway
Finished goods	STR (Stock Room)	Retail-ready products; vacuum/manual packing zone A

4.2 Production facilities (rooms)

Each facility is a physical area with one or more **production lines** and a linked **production warehouse**:

Facility	Production WH	Typical work
Oil Extraction	WH-PROD-OIL	Seed → oil extract → filter → fill variants
Milling Room	WH-PROD-MILL	Destone, hull, rice/ragi → semi-finished
Manual Cleaning	WH-PROD-MCLEAN	Grade/clean semi → bulk for STR
Flour Mill	WH-PROD-FLOUR	Flour, spice, ravva grinding
Soap Room	WH-PROD-SOAP	Soap process
Snacks Room	WH-PROD-SNACKS	Snacks production
Stock Room Packing	STR zone A	Vacuum and manual retail packing

4.3 Location hierarchy

Stock is always stored in **bins**:

```
Warehouse → Zone → Shelf → Bin
Example:   STR → A → S17 → 03
```

- **Godowns** are often scanned at **shelf** level (bin 00 = whole shelf).
- **Stock Room** uses full bin labels on every slot.

4.4 Key document types

Code prefix	Name	Meaning
PO-	Purchase order	Order to a vendor
GRN-	Goods receipt	Physical receipt against a PO
MO-	Manufacturing order	Production run
TRF-	Transfer order	Warehouse move (pick → drop)
SO-	Sales order	Customer order
INV-	Invoice	Bill to customer
TRP-	Trip	Truck dispatch plan

5. Desktop navigation

Left menu groups (top to bottom):

Overview: Dashboard

Catalog & buying: Products, Procurement, Price lists

Sales: Customers, Enquiries, Quotes, Sales orders, Picking, Packing, Returns

Stock: Inventory, Warehouse, Transfers

Production: Manufacturing, BOMs, Productivity

Logistics: Transport

Finance: Billing

Analytics: Reports, Container reports

Bottom: Approvals, Settings (admin)

6. Dashboard

Purpose: Plant-wide snapshot at login.

What you see:

- Production efficiency, active workers, delayed MOs
- Low-stock alerts (below reorder level)
- Sales vs COGS chart (14 days)
- Procurement spend by category
- Planned vs actual production
- Station load (work centres)
- Enquiry pipeline widget

Actions: Click KPIs or chart segments to drill into Manufacturing, Inventory, Enquiries, etc.

7. Products and catalog

Path: Products

Roles: Supervisor, Procurement, Admin

7.1 Product types

Type	Example	Used for
Raw	WSS (white sesame seeds)	Purchased inputs
Semi	GOIL-UNFILT	Intermediate manufacturing output
Finished	GOIL, RAGI	Sold products (bulk or packed)
Consumable	Packaging, chemicals	Used but not sold
Service	Labour, freight	Non-stock lines

7.2 Variants

A **parent product** (e.g. Groundnut Oil) can have **variants** (500 ml, 1 L, 5 L). Each variant has its own SKU, barcode, and stock counter.

7.3 Common tasks

Create a product

1. Click **New product**.
2. Enter SKU, name, type, UoM (kg, L, pc), category, prices.
3. Add variants if needed (size, pack size, barcode).
4. Save.

Edit / upload image

1. Select product in the list.
2. Use the detail panel → **Edit** or drag an image.

Supply outlook

Select a product → **Supply outlook** panel shows:

- On-hand stock
- Open purchase orders (incoming)
- Open manufacturing orders (pipeline)
- Stock rules and replenishment hints

Export

Use **Export CSV** for a spreadsheet of products or variants.

Find a SKU quickly

Use the search box above the list. If a SKU does not appear in dropdowns elsewhere, ensure it is active and the list limit includes it.

8. Procurement

Path: Procurement

Roles: Procurement, Admin

Three tabs: **PO**, **Vendors**, **GRN**.

8.1 Vendors

Add vendor: Vendors tab → **Add vendor** → code, name, GST, contact, payment terms, lead time.

Vendor catalog: Link vendor part numbers and pack sizes to internal SKUs (used for PO lines and auto-reorder).

Vendor detail: View performance metrics and catalog from the vendor row.

8.2 Purchase orders (PO)

Create PO

1. PO tab → **New PO**.
2. Select vendor, add lines (product, qty, rate).
3. Set expected delivery date.
4. Save as **draft**.

Approve PO

1. Open draft PO → **Approve**.
2. Status becomes **approved** — lines are locked (notes only).
3. Share PO with vendor via **Share** link if needed.

PO statuses

```
draft → approved → partial → received
                        ↘ closed (manual)
draft → cancelled (only if no GRNs)
```

Status	Meaning
draft	Being prepared
approved	Sent to vendor; awaiting delivery
partial	Some goods received
received	Fully received
closed	Manually closed (short-ship, write-off)
cancelled	Voided before receipt

8.3 Goods receipt (GRN)

Receive against PO (desktop)

1. Open approved PO → **Receive** (or GRN tab → receive from PO).
2. Enter accepted qty, rejected qty, batch/expiry if applicable.
3. **Allocate bins** — split qty across one or more destination bins (putaway).
4. Post GRN.

GRN QC

After receipt, QC status is set per GRN:

QC status	Meaning
pending	Awaiting inspection
pass	Accepted into stock
rework	Needs rework before use
reject	Rejected; qty not added to usable stock

Close PO

When all lines are received (or you accept short-ship), use **Close PO** with confirmation preview.

8.4 Procurement workflow summary

Need stock → Create PO → Approve → Vendor delivers
→ GRN receive + bin allocation → Stock in bins → PO received/closed

Stock rules can also **auto-draft POs** when global raw stock falls below minimum (Settings → Stock rules, trigger type **PO**).

9. Inventory and stock

Path: Inventory

Roles: Supervisor, Warehouse, Procurement, Admin

9.1 Tabs

Tab	Purpose
Locations	Find stock by product, warehouse, or bin; adjust qty
Ledger	Full audit trail of movements
Valuation	Inventory value snapshot
Batches	Lot/batch traceability (FIFO)

9.2 Finding stock

1. Locations tab → search by SKU, product name, or bin code.
2. Expand warehouse → zone → shelf → bin.
3. See qty, reserved qty, and product assignment.

9.3 Adjust stock

1. Select a bin or use **Adjust stock** from product context.
2. Choose **delta** (+/- qty) or **count** (set absolute qty).

3. Enter reason (physical match, damage, spillage, etc.).

4. Save — ledger row is written automatically.

Deep link from Manufacturing: When an MO shows a shortage, use the link to Inventory with `adjust=1` pre-filled.

9.4 Stock ledger transaction types

Type	Typical source
GRN	Goods receipt
Sale	Invoice / dispatch
Issue	MO material issue
Production	MO output / complete
Transfer	Transfer order pick/drop
Adjust	Manual adjustment or cycle count

9.5 Reserved quantity

Reserved qty on a bin means stock is earmarked for an open **pick list** or **in-transit transfer** but not yet finally consumed. Free stock = `qty - reservedQty`.

10. Warehouse and locations

Path: Warehouse

Roles: Supervisor, Warehouse, Admin

10.1 Tree navigation

Browse **Warehouse** → **Zone** → **Shelf** → **Bin**. Occupancy indicators (green / amber / red) show how full a bin is relative to capacity.

10.2 Bin operations

Action	When to use
Edit bin	Change capacity, notes
Bin layout	Bulk-create bins for a new zone/shelf
Rename/delete zone or shelf	Layout changes (must be empty first)
Bulk zone stock	Mass update qty across a zone

10.3 WH Audit

Path: WH Audit

Roles: Warehouse, Admin

Supervisor review of floor activity:

- **Variance tab** — cycle counts flagged as abnormal (>10% or >50 units difference)
- **Scans tab** — scan events where outcome was not OK

Use this to follow up with workers after mobile recounts.

11. Transfer orders

Path: Transfers

Roles: Supervisor, Warehouse, Admin

11.1 What is a transfer order?

A **transfer order (TO)** is a structured move: worker **picks** from source bin(s), carries stock, then **drops** at destination bin(s). Every pick and drop updates the stock ledger.

11.2 Transfer kinds

Kind	Created by	Typical flow
Replenishment	MO release or stock rule	Godown → production WH
Putaway	MO complete or GRN	Production WH / dock → STR or rule bin
Manual	User	Any warehouse-to-warehouse move

11.3 Transfer status flow

```
ready → [worker claims] → ready (assigned)
    → [pick confirmed] → in_transit
    → [drop confirmed] → done

Any non-done → cancel → cancelled
```

11.4 Desktop actions

- Filter by kind: All, Putaway, Replenishment, Manual
- Open detail slide-over to see lines, bins, linked MO
- **Cancel** (supervisor/admin) while draft or ready
- Assign worker (admin/supervisor)

11.5 Pick and drop rules

- Each line should have a **source bin** (`fromBinId`) before pick.

- If auto-created TO has no source bin, supervisor must assign one (or re-release MO after godown stock is seeded).
- On drop, destination bin may be auto-selected if empty bin exists in destination warehouse.

11.6 Putaway rules

Path: Settings → Putaway rules (or `/putaway-rules` , admin only)

Define: *When product X arrives, default destination = warehouse Y, bin Z.*

Used by GRN receive hints and MO complete putaway TOs.

12. Manufacturing

Path: Manufacturing, BOMs

Roles: Supervisor, Admin

See also: `docs/manufacturing-multi-step-bom.md` , `docs/soap-manufacturing-process.md`

12.1 Concepts

Term	Meaning
BOM	Bill of Materials — recipe (components, scrap, by-products, operations)
MO	Manufacturing Order — a planned production run
WO	Work Order — one shop-floor step (operation) on an MO
Release	Check materials at production WH; create replenishment TOs if short
Issue materials	Consume components from bins (FIFO lots)
Log output	Record good/scrap/rework qty mid-run
Complete	Post finished goods to inventory; close WOs; may create putaway TO

12.2 BOMs

Path: Manufacturing → BOMs (or `/manufacturing/boms`)

Create/edit BOM

1. Select product (and variant if variant-specific).
2. Add **components** with qty and UoM.
3. Add **operations** (sequence, line, machine, duration, QA flag) for multi-step processes.
4. Set default **facility** and **line**.
5. Set output qty per batch (e.g. 100 L per extract run).

BOM tools

- **Tree view** — nested sub-assemblies

- **Explode** — total raw materials for a planned qty
- **Where used** — which parent BOMs use this component

Pack BOMs are often auto-generated for retail variants (Rev-Pack revision).

12.3 Manufacturing orders

Create MO

1. Manufacturing → **New MO** (or Command Palette).
2. Select BOM, planned qty, facility/line, dates.
3. Save → status **planned**.

Release MO (planned only)

1. Select MO → **Release**.
2. System checks component stock at **production warehouse**.
3. If short → creates **replenishment TO** (one TO with all shortage lines).
4. If no shortage → MO moves to **in-progress**.

Issue materials

1. MO must be in-progress (or release with stock already at line).
2. **Issue materials** — consumes BOM explosion from bins; writes ledger.
3. First issue typically sets MO to **in-progress**.

Work orders (multi-step)

For BOMs with operations (oil extract/filter, grain mill/clean):

- Panel shows WOs per operation sequence.
- **Start** → **Log output** → **QA** (if required) → next operation.
- **Split operation** — divide work across parallel lines (e.g. six oil extractors).

Complete MO

1. All WOs complete (or single-step MO ready).
2. **Complete** (shortcut **F8**) — posts FG (+ by-products) to facility WH or putaway bin.
3. If putaway rule points to STR → auto **putaway TO** created.

12.4 MO status flow

```
planned → in-progress → [qc] → completed
  ↳ delayed (flag)
  ↳ cancelled
```

Status	Meaning
planned	Created; not yet released or issued
in-progress	Materials issued or production started
qc	Optional quality hold
completed	FG posted; WOs closed
delayed	Schedule flag (non-blocking)
cancelled	Voided

12.5 Oil extraction example (three BOMs)

1. **Extract BOM** — seeds → unfiltered semi + press cake
2. **Filter BOM** — unfiltered semi → bulk filtered oil
3. **Pack BOM** (auto) — bulk oil → retail variants (500 ml, 1 L, ...)

Each stage can be a separate MO linked to the appropriate facility line.

12.6 Grain milling example

1. **Mill MO** — raw grain → semi at WH-PROD-MILL
2. **Transfer** — semi mill → manual cleaning (stock rule or manual TO)
3. **Clean MO** — semi → bulk FG
4. **Putaway TO** — bulk → STR for retail packing

13. Sales, quotes, and fulfilment

Roles: Supervisor, Billing (varies by screen)

13.1 CRM — Enquiries

Path: Enquiries

Pipeline view: Drag cards between stages:

new → contacted → qualified → proposal → won / lost

Actions: Log activities, set follow-up tasks, convert to customer, mark lost (requires reason).

13.2 Quotes

Path: Quotes

1. **Create quote** → add customer, lines, validity date.
2. **Submit** → formal quote (revision history on edits).

3. **Accept** → creates sales order (or triggers **credit approval** if over limit).
4. **Share** → customer link for print/PDF view.

Quote statuses: draft, submitted, accepted, converted, expired, rejected

13.3 Sales orders

Path: Sales orders

Create from: Accepted quote, manual entry, or **ecommerce** (prepaid web order).

Fulfilment path (mandatory sequence):

Sales order → Pick list → Packing slip → Invoice

There is no shortcut to invoice without pick/pack (except ecommerce auto-flow).

SO statuses: confirmed, partially_invoiced, invoiced, on_hold, closed, cancelled

13.4 Picking

Path: Picking

Roles: Supervisor, Warehouse, Billing

1. Create or auto-generate pick list from SO.
2. **Auto-pick** — system allocates bins (may warn on shortfall).
3. Print pick list (/print/pick-list/:id) or assign to mobile worker.
4. Complete pick → status **picked** → can create packing slip.

Pick statuses: draft, picking, picked, cancelled

13.5 Packing

Path: Packing

1. Open packing slip from picked SO.
2. Enter **packed qty** per line (may differ from picked — billing uses packed qty).
3. **Multi-container mode** — split lines across boxes/crates with weights.
4. Mark **packed** → ready for invoice.

Pack statuses: open, packed, invoiced, cancelled

13.6 Customer payments

Path: Customers → select customer → **Record payment**

Payments allocate FIFO to open invoices. Advances reduce credit exposure on new quotes.

14. Billing and POS

Path: Billing

Roles: Billing, Admin

14.1 POS tab

Walk-in counter sales:

1. Search or scan products.
2. Add to cart; apply customer (optional) for price list.
3. Choose payment: cash, UPI, card, **credit** (requires customer with limit).
4. **Issue invoice** — stock decrements at invoice time.

14.2 Invoices tab

- List/filter invoices by status, date, customer.
- Open drawer for detail, payments, share link, dispatch assignment.
- Draw down remaining qty from open sales orders.

Invoice statuses: draft, issued, partial, paid, overdue

14.3 Keyboard shortcut

F2 or Command Palette → **New Invoice**

15. Transport and dispatch

Path: Transport

Roles: Supervisor, Warehouse

15.1 Trips

A **trip** is one truck + driver + date with multiple **invoice drops**.

Workflow

1. Create trip or **auto-schedule** for upcoming days.
2. Add invoices to trip (invoice picker).
3. **Start trip** → in_transit.
4. **Complete trip** when deliveries done.
5. **Cancel/reschedule** — rolls dispatches forward.

Trip statuses: scheduled, in_transit, completed, cancelled

Trips link to **Container reports** for manifest and throughput.

16. Returns and credit notes

Path: Returns

Roles: Supervisor, Billing, Warehouse

16.1 Workflow

1. **Import** return lines from Excel template (customer, SKU, qty, reason).

2. Manager **approves/rejects** each line.
3. **Finalize** → credit note + payment allocation.

Return statuses: pending_approval, processed, cancelled

Mobile: **Returns** screen for field processing where enabled.

17. Approvals

Path: Approvals

Roles: Supervisor, Billing

Central queue for policy gates:

Type	Example
Credit limit	Quote would exceed customer credit → approve to create SO
Stock adjustment	Large variance needs sign-off
PO amendment	Change to approved PO
Price override	Special pricing

Actions: Approve or Reject (reject requires reason).

Priority shown: low, med, high.

18. Reports

Path: Reports, Container reports

18.1 Reports hub

Grouped catalog: Inventory, Production, Procurement, Workforce, Financial.

Live charts on hub:

- Production trend (14 days)
- Procurement spend split
- Sales trend
- Station load

Use **Export** where available on individual report pages.

18.2 Container reports

Path: Container reports

Tab	Purpose
Pack manifest	Containers per packing slip
Item history	SKU/barcode → recent containers and trips
Trip manifest	All containers on a trip
Pack throughput	Daily slips/containers/weight

19. Settings and administration

Path: Settings

Roles: Admin only

Section	Configure
Company	Legal name, GSTIN, address, logo
Warehouses	Warehouse codes and kinds
Putaway rules	Product → destination bin
Stock rules	Auto MO / TO / PO triggers
Production facilities	Rooms, lines, machines
Dispatch options	Delivery methods
Packing	Defaults for packing slips
Container types	Box/crate definitions
Categories	Product categories
Users & roles	Accounts, PIN for mobile
Security	Password policies
Scanner	Barcode integration
SMS / Payment	Notifications, CCAvenue
Sync & backup	Offline sync, database backup
Appearance / Language	Branding, locale

19.1 Users and PIN

1. Settings → Users & roles.

2. Create user with username, password, **role**.
3. Set **6-digit PIN** for mobile login.
4. Worker role users use mobile only.

19.2 Stock rules (admin)

See Section 21.

20. Mobile warehouse app

URL: `/m/` (install as PWA on phone)

20.1 First login

1. Open mobile URL.
2. Login with PIN or password.
3. **Select warehouse** — device remembers choice (Tasks and scans are scoped to this WH).

20.2 Bottom tabs

Tab	Purpose
Tasks	Pick, pack, transfer work queue
Scan	Camera barcode → location or product
Count	Cycle count / bin recount
Verify	Flagged counts needing review
Profile	Punch in/out, switch warehouse, sign out

20.3 Tasks hub

Segments: Pick · Pack · Move (transfers) · More

- **My tasks** — work you claimed
- **Available** — unclaimed queue (refresh ~30s)

Claim a pick list, packing slip, or transfer before starting.

More shortcuts: GRN, GRN QC, Bulk zone, Returns

20.4 Pick (mobile)

1. Open claimed pick list.
2. Walk path — each line shows bin location.
3. Scan bin or confirm manually.
4. Enter qty picked.
5. Complete pick → may auto-create packing slip.

20.5 Pack (mobile)

1. Open packing slip.
2. Confirm packed qty per line.
3. **Multi-container:** assign lines to containers → seal → mark packed.

20.6 Transfer (mobile)

Two steps:

1. **Pick** — pull from source bin (must have `fromBin` assigned on line).
2. **Drop** — put away at destination (auto-suggests bin if not preset).

Statuses: ready → in_transit → done

Error “No source bin assigned”: Supervisor must assign source bin on desktop (Transfers detail) or re-release MO after godown stock exists.

20.7 GRN (mobile)

1. More → GRN → select open PO.
2. Enter received qty, rejects, batch.
3. Allocate to bins (with receive hints from putaway rules).
4. Post → optional QC queue.

20.8 GRN QC (mobile)

More → GRN QC → pending receipts → pass / rework / reject.

20.9 Count / bin detail

Scan or **Count** → scan bin:

- **Recount** — set new qty (variance flagged if large)
- **Reassign** — change product on bin
- **Quick adjust** — small correction

Reason codes: physical_match, damage, found_elsewhere, product_swap, spillage, expired, other.

20.10 Bulk zone

Mass-edit bins in a zone: filter zone/shelf, scan-to-row, update barcode and qty.

20.11 Profile and attendance

- **Punch in / out / break** — links to Productivity reports.
- **Switch warehouse** if you work across buildings.

21. Automation: stock rules

Path: Settings → Stock rules

Stock rules watch inventory and auto-create documents when stock falls below **minQty**.

21.1 Trigger types

Trigger	Creates	Monitors
mo	Manufacturing order	Bin qty (or effective stock incl. pipeline)
transfer	Replenishment TO	Monitor bin; pulls from source bin
po	Draft purchase order	Global product stock (no specific bin)

21.2 Key fields

- **minQty** — reorder point (ROP); trigger when effective stock < min
- **maxQty** — target level; MO/TO qty sized toward max
- **monitorBin** — bin whose qty is watched (MO/transfer triggers)
- **sourceBin** — where transfer pulls from (transfer triggers only)
- **bomId** — which BOM to use for auto MO

21.3 When rules run

- After **GRN, PO close, MO complete, transfer pick/drop, inventory adjust**
- **Periodic check** every 6 hours (server setting)
- Manual **Run all checks** (Settings → Stock rules)

21.4 Rule sets in this plant

Tag	Count (typical)	Behaviour
tally-jit	~126	Retail variant low in STR → pack MO
grain-milling	~61	Mill/clean staging low → mill or clean MO
raw-procurement	varies	Global raw low → draft PO

Important: A full rule scan can create **many MOs at once** if many monitor bins are below ROP. Open MOs count toward **effective** stock and prevent duplicate MOs for the same BOM.

21.5 Disabling auto MOs (testing)

- Deactivate specific rules in Settings.
- Set env `STOCK_RULES_CHECK_INTERVAL_MS=0` to disable periodic scan (server admin).
- Avoid **Check all stock rules** until staging bins are stocked.

22. End-to-end scenarios

22.1 Buy raw material and put in godown

Procurement: Create PO → Approve
GRN: Receive → allocate to WH-STOR bin
Stock visible in Inventory → Locations

22.2 Make oil (full chain)

Manufacturing: Create MO (extract BOM) → Release
→ Replenishment TO: seeds godown → WH-PROD-OIL (mobile pick/drop)
→ Issue materials → Run extract WO → Log output
→ Complete → semi on WH-PROD-OIL
Filter MO → issue semi → filter WO → complete → bulk oil
Pack MO (or tally-jit auto) → fill variants → STR bins
Putaway TO if FG lands on production WH first

22.3 Sell to shop customer

Billing POS: Scan products → Pay → Invoice (stock decrements)

22.4 Sell to dealer (order fulfilment)

Quote → Accept → Sales order
Picking: Auto-pick → mobile pick
Packing: Pack → containers
Billing: Invoice from SO
Transport: Add invoice to trip → dispatch

22.5 Ecommerce prepaid order

Customer pays on web store
→ Confirmed SO + invoice + draft pick list (automated)
Warehouse: Pick → Pack → dispatch

22.6 Replenish retail shelf (JIT)

STR variant bin drops below ROP (tally-jit rule)
→ Auto pack MO OR transfer from bulk bin (if rule configured)
Complete MO → FG to STR

23. Glossary

Term	Definition
STR	Stock Room — finished-goods warehouse
Godown	Bulk raw storage (Big Godown WH-STOR, New Godown WH-GDNW)
Bin	Smallest stock location (zone/shelf/bin)
FIFO lot	Oldest receipt batch consumed first on issue
Effective stock	On-hand + open PO pipeline + open MO pipeline
Replenishment TO	Move raw from storage to production WH for an MO
Putaway TO	Move FG from production WH to STR (or rule destination)
WIP	Work in progress on facility production warehouse
Semi / Semi-FG	Intermediate product between process steps
By-product	Secondary output (e.g. press cake from oil extraction)
ATP	Available-to-promise — stock check at quote time
ROP	Reorder point — minQty on stock rules
GRN	Goods Receipt Note
MO / WO / TO / PO / SO	Manufacturing / Work / Transfer / Purchase / Sales order

24. Tips and troubleshooting

24.1 Common issues

Problem	Likely cause	What to do
MO release creates TO but pick fails	No source bin on TO line	Assign bin on Transfers detail, or seed godown stock and re-release
Many MOs appeared at once	Stock rules batch scan	Review planned MOs; deactivate rules if testing; cancel unneeded MOs
SKU missing in dropdown	List limit or inactive product	Search in Products; use Find SKU on Inventory adjust
Pick list shortfall (409)	Insufficient free bin qty	Check reserved qty; complete or cancel other picks
Export button does nothing	Browser blocked download	Allow downloads; check console; use CSV export buttons on each page
Mobile "No source bin"	TO line missing fromBinId	Supervisor assigns on desktop or fix replenishment config

24.2 Keyboard shortcuts (desktop)

Shortcut	Action
Ctrl+K / Ctrl+/	Command Palette
Ctrl+B	Scanner overlay
F2	New invoice (Billing)
F8	Complete MO (Manufacturing)

24.3 Related technical docs

Document	Topic
docs/manufacturing-multi-step-bom.md	Multi-operation BOMs
docs/soap-manufacturing-process.md	Soap room process
docs/warehouse-location-barcodes.md	Barcode labels and scanning
docs/manufacturing-inventory-storage.md	Storage model detail

24.4 Training progression (recommended)

1. **Week 1 — Stock:** Products, Inventory locations, Warehouse tree, mobile Scan + Count
2. **Week 2 — Inbound:** Procurement PO → GRN → putaway (desktop + mobile GRN)

3. **Week 3 — Moves:** Transfers, mobile pick/drop, putaway rules
 4. **Week 4 — Production:** BOMs, MO release/issue/complete, work orders
 5. **Week 5 — Outbound:** Quotes, SO, pick, pack, invoice, trips
 6. **Week 6 — Admin:** Settings, stock rules, users, approvals
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Document maintained for PVS ERP Portal. For deployment and server setup, see `DEPLOYMENT.md` in the repository root.